Machine-checked identifiability and exact degeneracy theorems for exoplanet transmission spectra

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Abstract

Which atmospheric parameters a noiseless transmission spectrum $R_p(\lambda)$ can determine, and which are linked by exact degeneracies, is usually studied numerically, through the shapes of posterior distributions. We instead formalise two forward operators in the Lean 4 proof assistant and prove identifiability and non-identifiability theorems about them. All results are checked by Lean's kernel and use only its standard axioms. The first operator is an exact geometric model: a hydrostatic atmosphere with arbitrary temperature profile $T(r)$ and arbitrary mixing-ratio profiles $X_i(r)$, observed through the exact chord integral. In this model, with temperature-independent cross-sections and a spectrally inactive filler gas, we prove that *any* temperature profile can be traded exactly for compensating mixing ratios. An explicit consequence is that a normal and a thermally inverted atmosphere can have identical spectra. We also prove the converse: closing the composition removes this freedom. The second operator is the isothermal analytic formula of Heng and Kitzmann [7]. For it we describe every fibre of the forward map for one absorber plus a grey cloud: exactly three of seven parameters are identifiable, and the rest are exact gauge directions. We also formalise the grey-cloud and power-law degeneracies [8]. For several absorbers, we reduce scale-height identifiability to a property of the opacity cone, and give a constructive counterexample showing that distinct cross-sections do not suffice. Finally, we prove that the neglected $E_1(\tau_0)$ term breaks the reference-radius/pressure degeneracy for every admissible model. Attempting proofs of three open targets also exposed two subtly wrong problem statements, which we document.

1 Introduction

Retrievals of exoplanet transmission spectra routinely run into degeneracies: between the reference radius and the reference pressure, between abundances and that pressure, between temperature and gravity, and between clouds and molecular features [1, 6, 7, 10]. In practice these appear as elongated or curved posterior distributions in Markov-chain Monte Carlo runs. Such numerical evidence cannot separate an *exact* degeneracy of the forward model from a near-degeneracy caused by finite noise, limited wavelength coverage or a poorly mixing sampler.

The underlying question is one of *structural identifiability*. Given a forward map $F : \Theta \to (\lambda \mapsto R_p(\lambda))$, what are its fibres $\{\theta' : F(\theta') = F(\theta)\}$? An exact degeneracy is a non-trivial fibre. The most convincing evidence for one is an explicit second parameter set with exactly the same image. An identifiability theorem is a proof that the fibres are no larger than a known symmetry group.

Classical analytic formulas for $R_p(\lambda)$ [2, 4, 5, 7, 8] rest on a stack of explicit assumptions: isothermal, isochemical, constant gravity, grey or power-law opacities, and the optically thick limit. Each of these changes the fibre structure. We make every assumption an explicit hypothesis and verify every argument mechanically in Lean 4 [3] with Mathlib [9]. A proof assistant guards against two errors that are easy to make in this area: proving a theorem under a hidden assumption, and stating a conjecture that is vacuously true or trivially false. We met the second kind ourselves (Section 9).

Contributions.

1. An exact, non-isothermal, non-isochemical forward model, with a proof that the spectrum depends on the atmosphere only through the number densities of the active species (Theorem 2).

2. An exact temperature–abundance profile degeneracy: any $T(r)$ is reproducible (Theorem 3). It includes a constructive normal/inverted twin pair (Theorem 6), and a converse showing exactly which assumption the degeneracy rests on (Theorem 7).

3. A complete fibre description of the Heng and Kitzmann [7] operator for one absorber plus a grey cloud (Theorem 11), together with its gauge group (Proposition 9) and the grey and power-law degeneracies (Section 4.1).

4. A reduction of multi-absorber identifiability to *power rigidity* of the opacity cone (Theorem 17), a constructive counterexample (Theorem 18), and a proof of rigidity for one band (Theorem 19).

5. A proof that the $E_1(\tau_0)$ correction breaks the (R_0, P_0) gauge for every model (Theorem 20), and a resolution of a density-identifiability target (Section 7).

The development has 14 Lean files, about 2,000 lines, 83 theorems and 44 definitions. It is available at `https://github.com/curiousug11/exo_lean`; Table 2 maps each result to its Lean name.

2 Forward models and the assumption lattice

2.1 The exact model

An atmosphere $\mathcal{A} = (R_0, P_0, g, m, T, X)$ consists of a reference radius R_0, the pressure P_0 at R_0, a constant gravitational acceleration g, a constant mean particle mass m, a temperature profile $T : \mathbb{R} \to \mathbb{R}$, and mixing-ratio profiles $X_i : \mathbb{R} \to \mathbb{R}$ for the spectrally active species $i \in \iota$. Neither T nor X_i is assumed constant. The pressure is *defined* as the solution of hydrostatic equilibrium,

$$P(r) = P_0 \exp\Big(- \int_{R_0}^{r} \frac{mg}{k_{\mathrm{B}} T(s)} \, \mathrm{d}s \Big), \qquad n(r) = \frac{P(r)}{k_{\mathrm{B}} T(r)}, \qquad n_i(r) = X_i(r)\, n(r), \tag{1}$$

and we prove that $P'(r) = -\rho(r)\, g$ with $\rho = mn$, whenever $\mathcal{A}$ is *well posed*: $P_0, g, m > 0$, and T continuous and positive (`hasDerivAt_pressure`). With cross-sections $\sigma_i(\lambda)$ independent of temperature and pressure, the extinction coefficient is $\alpha(r, \lambda) = \sum_i \sigma_i(\lambda) n_i(r)$. The transit radius is the exact geometric chord integral, without refraction, for an opaque core of radius R_0 and an atmosphere truncated at R_{top}:

$$\tau(b, \lambda) = 2 \int_{b}^{R_{\mathrm{top}}} \frac{\alpha(r, \lambda)\, r}{\sqrt{r^2 - b^2}} \, \mathrm{d}r, \qquad R_p(\lambda)^2 = R_0^2 + 2 \int_{R_0}^{R_{\mathrm{top}}} b \left(1 - e^{-\tau(b,\lambda)}\right) \mathrm{d}b. \tag{2}$$

The gas not accounted for by the listed species $(1 - \sum_i X_i)$ is a transparent *filler*. Mixing ratios are *admissible* above R_0 if $X_i \geq 0$ and $\sum_i X_i \leq 1$.

Assumption	Exact layer	Analytic layer
constant g, constant m	structural	structural
T, P-independent cross-sections	structural	structural
transparent filler gas, no refraction	structural	structural
isothermal / isochemical	**not assumed**	structural
optically thick limit ($E_1(\tau_0) \to 0$)	not needed	structural (relaxed in Section 6)
grey / power-law opacity	hypothesis	hypothesis

Table 1: The assumption lattice (`ExoLean/Assumptions.lean`).

2.2 The analytic model

In the isothermal, isochemical regime and the optically thick limit, Heng and Kitzmann [7] give

$$R_p(\lambda) = R_0 + H\Big[\gamma + \ln\Big(\frac{P_0\,\kappa(\lambda)}{g}\sqrt{\frac{2\pi R_0}{H}}\Big)\Big], \quad H = \frac{k_{\mathrm{B}}T}{mg}, \quad \kappa(\lambda) = \sum_i a_i\sigma_i(\lambda) + \kappa_c, \quad a_i = \frac{X_i}{m}, \tag{3}$$

where γ is the Euler–Mascheroni constant and κ_c is a grey cloud opacity. We take (3) as the *definition* of the analytic operator. Its asymptotic derivation from (2) is not formalised. A parameter set $\theta = (T, m, g, R_0, P_0, X, \kappa_c)$ is *valid* if $T, m, g, R_0, P_0 > 0$ and $X_i, \kappa_c \geq 0$. We define the *level* $L(\theta) = R_0 + H[\gamma + \ln(P_0 g^{-1}\sqrt{2\pi R_0/H})]$.

2.3 The assumption lattice

Table 1 lists which assumptions each layer uses. Structural assumptions are built into a model. Regime assumptions are hypotheses that a theorem takes explicitly. The two layers are connected by a bridge lemma.

Lemma 1 (bridge; `speciesDensity_of_isothermal_isochemical`)**.** *If $T \equiv T_0$ and $X_i \equiv x_i$, then $n_i(r) = x_i P_0 (k_{\mathrm{B}}T_0)^{-1}\exp(-(r - R_0)/H)$ with $H = k_{\mathrm{B}}T_0/(mg)$.*

3 Exact layer: non-isothermal profile degeneracies

Theorem 2 (factorisation; `spectrum_eq_of_speciesDensity_eq`)**.** *If two atmospheres share R_0 and $n_i^{\mathcal{A}}(r) = n_i^{\mathcal{B}}(r)$ for all i, r, then $R_p^{\mathcal{A}} = R_p^{\mathcal{B}}$ for every set of cross-sections and every R_{top}.*

The spectrum therefore never sees T, X, P, g or m separately. The question becomes which changes of (T, X) preserve every n_i while keeping the atmosphere hydrostatic.

Theorem 3 (temperature profile is unidentifiable; `temperature_profile_unidentifiable`)**.** *Let $\mathcal{A}$ be well posed and let $T' : \mathbb{R} \to (0, \infty)$ be any continuous profile. Let P' be the hydrostatic pressure of T' (same R_0, P_0, g, m) and set*

$$X_i'(r) = X_i(r)\,\frac{T'(r)}{T(r)}\,\frac{P(r)}{P'(r)}. \tag{4}$$

Then $\mathcal{A}' = (R_0, P_0, g, m, T', X')$ is well posed and hydrostatic, $n_i' \equiv n_i$, and $R_p^{\mathcal{A}'}(\lambda) = R_p^{\mathcal{A}}(\lambda)$ for all λ, all cross-sections and all R_{top}.

Proof sketch. $n_i' = X_i' P'/(k_{\mathrm{B}} T') = X_i P/(k_{\mathrm{B}} T) = n_i$ by (4). The hydrostatic property follows from the fundamental theorem of calculus for continuous $1/T'$. Conclude with Theorem 2. □

The compensating mixing ratios must still be admissible.

Proposition 4 (admissibility; `PhysicalVMR.retemper`)**.** *Suppose the mixing ratios of $\mathcal{A}$ are admissible, and for $r \geq R_0$ we have $T(r) \leq T'(r) \leq cT(r)$ and $c\sum_i X_i(r) \leq 1$. Then the mixing ratios of $\mathcal{A}'$ are admissible. The proof uses that a hotter atmosphere is more extended: $P \leq P'$ above R_0 (`pressure_le_withT`), so $0 \leq X_i' \leq cX_i$.*

Corollary 5 (isothermal twin; `isothermal_twin`)**.** *An isothermal, isochemical atmosphere at T_0 with mixing ratios x_i has the same exact spectrum as the isothermal atmosphere at cT_0 ($c > 0$) with $X_i'(r) = c\, x_i \exp[-(1 - c^{-1})(r - R_0)/H]$. If $c \neq 1$ and $x_i \neq 0$, then X_i' is not constant.*

Corollary 5 pins down the role of the isochemical assumption in reading a temperature off the scale height. Only the *absorber* scale height H_a is observable, with $1/H_a = 1/H + 1/H_X$; an upward-depleted absorber in a hotter atmosphere mimics a cooler, well-mixed one.

Theorem 6 (normal and inverted twins; `exists_inverted_twin`)**.** *Let $s(x) = 1/(1 + e^{-x})$. Fix $R_0, P_0, g, m > 0$, $T_0 > 0$, $0 < \Delta T \leq T_0$, $L > 0$, and constant mixing ratios $x_i \geq 0$ with $2\sum_i x_i \leq 1$. Set*

$$T_{\mathrm{n}}(r) = T_0 + \Delta T\, s\big(-\tfrac{r - R_0}{L}\big), \qquad T_{\mathrm{inv}}(r) = T_0 + \Delta T\, s\big(\tfrac{r - R_0}{L}\big).$$

Let $\mathcal{A}$ have profile T_{n} and mixing ratios x, and let $\mathcal{A}'$ be its re-tempered twin with profile T_{inv}. Then both are well posed with admissible mixing ratios. T_{n} is strictly decreasing outward, T_{inv} is strictly increasing (a thermal inversion), neither atmosphere is isothermal, and their exact spectra coincide for all cross-sections and all R_{top}.

This degeneracy rests on two assumptions: temperature-independent cross-sections, and a filler gas the spectrum never sees. The next theorem shows the second one is essential.

Theorem 7 (closing the composition; `T_eq_of_speciesDensity_eq`)**.** *Let $\mathcal{A}, \mathcal{B}$ be well posed with the same R_0, P_0, g, m, and suppose $\sum_i X_i = 1$ on $[R_0, R_{\mathrm{top}}]$ for both (every constituent is listed). If $n_i^{\mathcal{A}} = n_i^{\mathcal{B}}$ on $[R_0, R_{\mathrm{top}}]$ for all i, then $T^{\mathcal{A}} = T^{\mathcal{B}}$ on $[R_0, R_{\mathrm{top}}]$.*

Proof sketch. Equal species densities give equal total densities, so $\frac{\mathrm{d}}{\mathrm{d}r}(P^{\mathcal{A}} - P^{\mathcal{B}}) = 0$ on the column. The difference vanishes at R_0, so the pressures agree, and then $T = P/(k_{\mathrm{B}} n)$ agrees. □

Physically, collision-induced absorption or Rayleigh scattering by an H_2/He background makes the filler visible. That breaks Theorem 3, consistent with the important role of collision-induced absorption found in numerical retrievals [10].

4 Analytic layer: fibres and gauge symmetries

Lemma 8 (structural identity; `Iso.transitRadius_eq`)**.** *For valid θ and $\kappa(\lambda) > 0$: $R_p(\lambda) = L(\theta) + H \ln \kappa(\lambda)$.*

Proposition 9 (exact gauge symmetries)**.** *For valid θ and $c > 0$, each of the following leaves $R_p(\lambda)$ unchanged:*

(i) the reference gauge $(R_0, P_0) \mapsto (R_0',\, P_0\sqrt{R_0/R_0'}\, e^{(R_0 - R_0')/H})$ for any $R_0' > 0$, at every λ with $\kappa(\lambda) > 0$. The $\sqrt{R_0}$ factor of (3) is compensated exactly (`Iso.transitRadius_referenceGauge`);

(ii) normalisation $(P_0, X_i, \kappa_c) \mapsto (P_0/c,\, cX_i,\, c\kappa_c)$ *[1, 7]* (`Iso.transitRadius_normalization`);

(iii) temperature–gravity $(T, g, P_0) \mapsto (cT, cg, cP_0)$ (`Iso.transitRadius_T_g`);

(iv) temperature–mean mass $(T, m, X_i) \mapsto (cT, cm, cX_i)$ (`Iso.transitRadius_T_m`).

All the regime degeneracies below follow from one engine: any target level can be reached by tuning P_0 alone (`Iso.exists_P0_level`). The converse to Proposition 9 rests on a rigidity lemma.

Lemma 10 (log-affine rigidity; `logAffine_rigidity`)**.** *Let* $a, a' > 0$, $\kappa, \kappa' \geq 0$, $H > 0$, *and* $0 \leq u_1 < u_2$. *If* $L + H\ln(au + \kappa) = L' + H'\ln(a'u + \kappa')$ *for all* $u \in (u_1, u_2)$, *then* $H = H'$, $a\kappa' = a'\kappa$ *and* $L = L' + H\ln(a'/a)$.

Proof sketch. Differentiating on the open interval gives $Ha(a'u + \kappa') = H'a'(au + \kappa)$, a linear identity in u. Comparing its two coefficients gives the first two relations, and evaluating at one point gives the third. □

Theorem 11 (fibres of the analytic map; `Iso.transitRadius_fibre`)**.** *Consider one absorber with cross-section* $\sigma > 0$ *on a wavelength set* S, *such that* $\sigma(S)$ *contains an interval* (u_1, u_2) *with* $u_1 \geq 0$, *plus a grey cloud. For valid* θ, θ' *with* $X, X' > 0$,

$$R_p^{\theta} = R_p^{\theta'} \text{ on } S \iff H = H', \quad a\,\kappa_c' = a'\kappa_c, \quad L(\theta) = L(\theta') + H\ln(a'/a).$$

The identifiable coordinates are therefore $(H,\ \kappa_c/a,\ L + H\ln a)$: three numbers out of the seven parameters $(T, m, g, R_0, P_0, X, \kappa_c)$. The four-dimensional fibres are exactly the orbits of Proposition 9.

Corollary 12 (`Iso.temperature_identifiable`, `Iso.cloudRatio_identifiable`, `Iso.radius_identifiable_of_P0`)**.** *Under the hypotheses of Theorem 11: T is identifiable if m and g are known; the cloud-to-absorber ratio κ_c/a is identifiable. With all other parameters fixed (including P_0), $R_p(\lambda)$ is strictly increasing in R_0, so a single wavelength determines R_0.*

4.1 Opacity regimes

Proposition 13 (grey degeneracy; `Iso.grey_temperature_unidentifiable`)**.** *If every σ_i is constant on S and $\kappa > 0$ there, then for every $T' > 0$ there is a $P_0' > 0$ such that θ with (T, P_0) replaced by (T', P_0') has the same spectrum on S. A flat spectrum leaves the temperature completely unconstrained.*

Theorem 14 (power-law slope [8]; `Iso.lecavelier_slope`)**.** *If* $\kappa_c = 0$, $\sigma_i(\lambda) = K_i\lambda^{\beta}$ *and* $\sum_i a_i K_i > 0$, *then* $\mathrm{d}R_p/\mathrm{d}\ln\lambda = \beta H$ *exactly.*

Proposition 15 (slope–temperature degeneracy; `Iso.powerLaw_slope_degeneracy`, `Iso.slopeProduct_identifiable`)**.** *If the slope is a free parameter, then for every $c > 0$ the model at $(cT, \beta/c)$, with P_0 re-tuned, reproduces the model at (T, β) at every $\lambda > 0$. Conversely, agreement at two distinct wavelengths forces $\beta H = \beta' H'$, so only the product is identifiable.*

5 Several absorbers: power rigidity

Let $\mathcal{C}(\sigma) = \{\sum_i a_i\sigma_i + c : a_i, c \geq 0\}$ be the opacity cone. If $R_p^{\theta} = R_p^{\theta'}$ on S, the structural identity gives $\kappa_{\theta'} = C\,\kappa_{\theta}^{r}$ on S, with $r = H/H' > 0$ and $C = e^{(L-L')/H'} > 0$.

Definition 16 (`PowerRigid`)**.** σ is *power-rigid* on S if, whenever $\kappa, \kappa' \in \mathcal{C}(\sigma)$ with $\kappa > 0$ on S, κ non-constant on S, $r > 0$, $C > 0$ and $\kappa' = C\kappa^r$ on S, it follows that $r = 1$.

Theorem 17 (reduction; `Iso.scaleHeight_eq_of_powerRigid`)**.** *If σ is power-rigid on S, both opacities are positive on S, and the spectrum is not flat, then equal spectra on S imply $H = H'$.*

Theorem 18 (power-related twins; `Iso.exists_powerRelated_twin`, `Iso.rayleigh_lambda8_twin`)**.** *Suppose $\sigma_1 = \sigma_0^\rho$ on S for some $\rho > 0$, with $\sigma_0 > 0$ there. Then a cloud-free atmosphere containing only species 0 at temperature T and one containing only species 1 at T/ρ (same m, g, R_0; re-tuned P_0) have identical spectra on S. In particular, a pure λ^{-4} Rayleigh scatterer at T is indistinguishable from a pure λ^{-8} absorber at $T/2$, and $\{\lambda^{-4}, \lambda^{-8}\}$ is not power-rigid (`Iso.not_powerRigid_rayleigh_lambda8`).*

Theorem 18 refutes the intuition that distinct (even linearly independent) cross-sections suffice for identifiability. What matters is the absence of multiplicative relations within the cone.

Theorem 19 (single-band rigidity; `powerRigid_of_unique`, `gaussianBandsPowerRigid_of_unique`)**.** *Any single cross-section that is continuous and positive on an open band, together with a grey cloud, is power-rigid on that band. In particular, every single Gaussian band* $\exp(-((\lambda - c)/s)^2)$ *is.*

Proof sketch. By the intermediate value theorem, σ maps the band onto a set containing an interval of positive values. Taking logarithms of $\kappa' = C\kappa^r$ turns the identity into the form of Lemma 10 with $H = r$, $H' = 1$, which forces $r = 1$. □

Power rigidity of two or more Gaussian bands with distinct centres remains open (Section 10).

6 Beyond the optically thick limit

The full formula of Heng and Kitzmann [7] is $R_p = R_0 + H[\gamma + E_1(\tau_0) + \ln \tau_0]$, with $\tau_0 = P_0 \kappa g^{-1} \sqrt{2\pi R_0/H}$ and $E_1(\tau) = \int_1^\infty e^{-\tau t} t^{-1} \mathrm{d}t$.

Theorem 20 (E_1 breaks the reference gauge; `Iso.transitRadiusE1_referenceGauge`, `Iso.transitRadiusE1_gauge_ne`, `Iso.transitRadiusE1_gauge_lt`)**.** *For valid θ, $\kappa(\lambda) > 0$ and $R_0' > 0$, the gauge transformation of Proposition 9(i) maps $\tau_0 \mapsto \tau_0 e^{(R_0 - R_0')/H}$ and shifts the transit radius by exactly*

$$\Delta R_p = H\left[E_1\left(\tau_0 e^{(R_0 - R_0')/H}\right) - E_1(\tau_0)\right].$$

E_1 is strictly decreasing on $(0, \infty)$ (`expIntegralE1_strictAntiOn`), so $\Delta R_p \neq 0$ whenever $R_0' \neq R_0$, and $\Delta R_p > 0$ whenever $R_0' > R_0$.

Since $E_1(\tau) \leq e^{-\tau}/\tau$ (`expIntegralE1_le`), the breaking is exponentially small for $\tau_0 \gg 1$. The (R_0, P_0) degeneracy is therefore exact only in the limit, and effectively exact in the optically thick regime.

7 Do spectra determine density profiles?

By Theorem 7, identifiability of $T(r)$ in a closed-composition atmosphere reduces to whether the spectrum determines the $n_i(r)$ on $[R_0, R_{\mathrm{top}}]$.

Proposition 21 (`not_densityProfilesIdentifiable`)**.** *Without further restrictions the answer is no, for every cross-section set. A zero-thickness column ($R_0 = R_{\mathrm{top}}$) has $R_p \equiv R_0$ whatever the atmosphere. Changing a discontinuous mixing ratio at a single radius is another failure, invisible to every integral.*

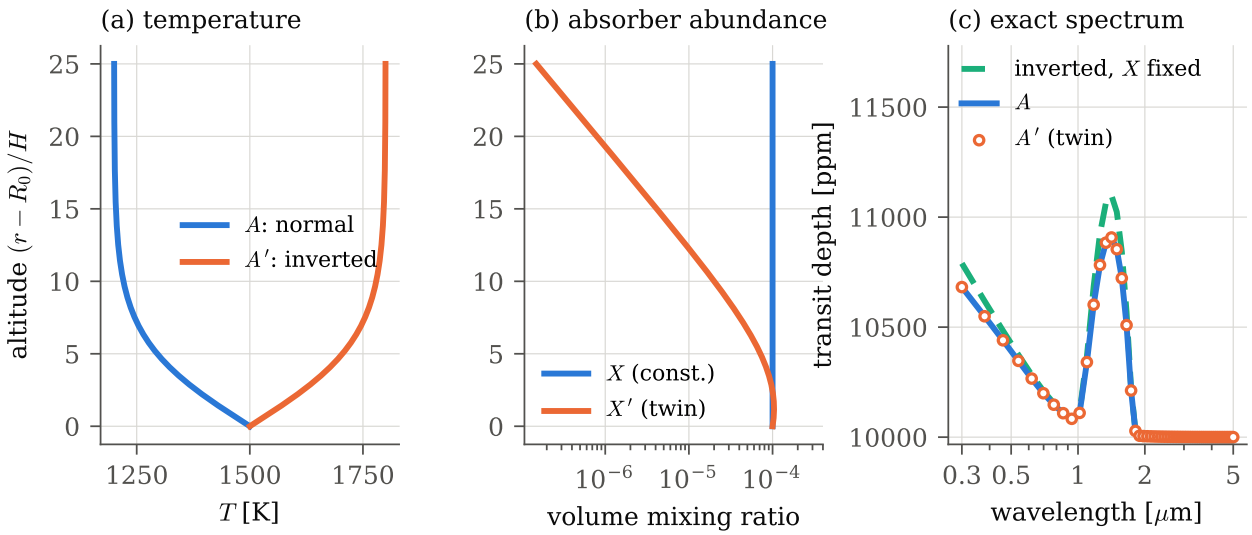


Figure 1: The normal/inverted twin of Theorem 6 ($T_0 = 1200\,\mathrm{K}$, $\Delta T = 600\,\mathrm{K}$, $L = 3H$). (a) Temperature profiles. (b) Constant mixing ratio of the normal atmosphere and compensating profile (4) of the inverted one. (c) Exact transit depths. The twin (open circles) coincides with $\mathcal{A}$ to floating-point precision. Applying the inversion without the compensation (dashed) changes the depth by up to 190 ppm.

We therefore restrict to $R_0 < R_{\mathrm{top}}$ and continuous X_i (`DensityProfilesIdentifiableCts`).

Theorem 22 (proportional cross-sections; `not_densityProfilesIdentifiableCts_of_proportional`). *If $\sigma_j = k\sigma_i$ on S for some $i \neq j$ and $k > 0$, the corrected target fails. The atmosphere with $X_i = k/(1+k)$ and the one with $X_j = 1/(1+k)$ (all else equal, all profiles constant and admissible) have identical spectra on S but different density profiles.*

Theorem 23 (conditional identifiability; `temperature_identifiable_of_densityIdentifiable`). *If the corrected target holds for σ, then for closed, continuous compositions with shared R_0, P_0, g, m and $R_0 < R_{\mathrm{top}}$, the spectrum determines $T(r)$ on $[R_0, R_{\mathrm{top}}]$.*

For a single species, (2) reads $R_p^2 - R_0^2 = 2\int b\,(1 - e^{-\sigma N(b)})\,\mathrm{d}b$, where $N(b)$ is the Abel-transformed column density. As a function of σ this is a Laplace-type transform of the *distribution* of N. So uniqueness can hold at best up to rearrangement of N, for example when N is monotone.

8 Numerical cross-check

To check that the Lean definitions mean what we intend, we evaluate (2) and (3) numerically (`scripts/numerical_check.py`). The numerical results are not part of any proof. The test planet has $R_0 = 7 \times 10^7\,\mathrm{m}$, $g = 10\,\mathrm{m\,s^{-2}}$, $\mu = 2.3$, $P_0 = 1\,\mathrm{bar}$ and $H \approx 434\,\mathrm{km}$. It has one absorber with $\sigma(\lambda) = 10^{-27}(\lambda/0.5\,\mu\mathrm{m})^{-4} + 4 \times 10^{-26} e^{-((\lambda - 1.4\,\mu\mathrm{m})/0.15\,\mu\mathrm{m})^2}\,\mathrm{m}^2$ and $X = 10^{-4}$, and the atmosphere is truncated at $R_{\mathrm{top}} = R_0 + 25H$. Transit depths are $(R_p/R_\star)^2$ with $R_\star = 7 \times 10^8\,\mathrm{m}$.

Figure 1 illustrates Theorem 6. The compensating mixing ratio stays in $[1.5 \times 10^{-7}, 1.04 \times 10^{-4}]$, within the bound $X' \leq 2X$ of Proposition 4. The maximum depth difference between the twins is 0 to floating-point precision. Without compensation, the inversion changes the depth by up to 190.5 ppm. Corollary 5 with $c = 1.5$ reproduces the depths to 2×10^{-11} ppm. Figure 2 shows that

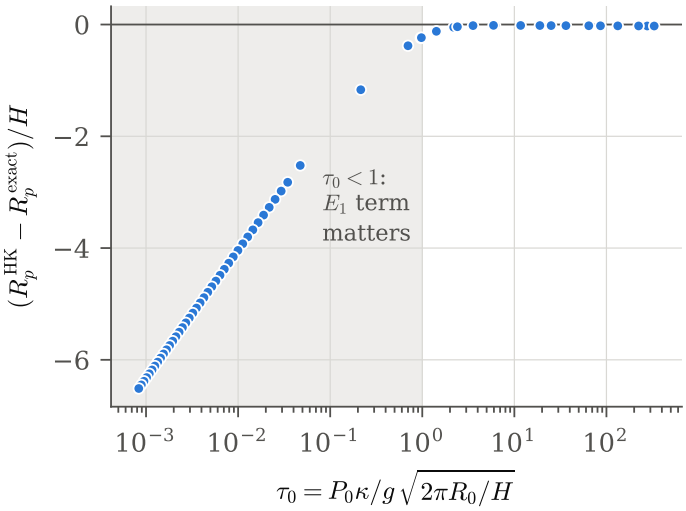


Figure 2: Analytic formula (3) minus the exact integral (2), in units of H, for the isothermal model, against τ_0. For $\tau_0 > 10$ the two agree within $0.03H$. For $\tau_0 < 1$ (shaded), the neglected $E_1(\tau_0)$ term and the opaque lower boundary dominate.

```
theorem temperature_profile_unidentifiable (hA : A.WellPosed) (T' : ℝ → ℝ)
    (hc : Continuous T') (hT' : ∀ r, 0 < T' r) :
    ∃ A' : Atmosphere ι, A'.WellPosed ∧ A'.T = T' ∧
      A'.R0 = A.R0 ∧ A'.P0 = A.P0 ∧ A'.g = A.g ∧ A'.m = A.m ∧
      (∀ i r, A'.speciesDensity i r = A.speciesDensity i r) ∧
      ∀ σ Rtop, A'.spectrum σ Rtop = A.spectrum σ Rtop
```

Figure 3: Lean statement of Theorem 3 (`ExoLean/ProfileDegeneracy.lean`).

the analytic formula agrees with the exact integral to $0.027H$ at all 10 of the 60 wavelengths with $\tau_0 > 10$. Where $\tau_0 < 1$ it falls up to $6.5H$ below, which is the regime of Theorem 20.

9 Formalisation notes

The development uses Lean 4 (v4.35.0) and Mathlib (v4.35.0). All headline theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`, as checked by `scripts/CheckAxioms.lean`, and no `sorry` remains. Figure 3 shows the Lean statement of Theorem 3. Analytic ingredients come from Mathlib: the fundamental theorem of calculus for hydrostatics, uniqueness of derivatives for Lemma 10, the mean value theorem for Theorem 7, the intermediate value theorem for Theorem 19, and Bochner integration on $(1, \infty)$ for E_1.

Specification errors found by attempting proofs. Two of our own initial problem statements were wrong, and the proof attempts exposed both:

1. *Power rigidity* was first stated without $r, C > 0$. With $r = 0$, any constant cloud satisfies $\kappa' = C\kappa^0$, so no family with a non-constant cone element could be rigid (`not_powerRigidUnrestricted`). The reduction theorem was true but vacuous. Since the physics only produces $r = H/H' > 0$ and $C > 0$, the corrected definition loses nothing.

2. *Density identifiability* was first stated over all atmospheres. That class admits zero-thickness columns and pointwise modifications, which made the target false for trivial reasons (`not_densityProfilesIdentifiable`).

In both cases the error did not show up in prose, and was found when a proof was attempted. Identifiability conjectures are statements about *all* parameter sets, so a single overlooked edge case can make them vacuous or trivially false. We see this as an argument for stating such conjectures formally.

10 Open problems

1. **Multi-band power rigidity** (`GaussianBandsPowerRigid`, $|\iota| \geq 2$). A plausible route is in three steps. First, extend $g = Cf^r$ from the band to $\mathbb{R}$ by real analyticity, since $f > 0$ everywhere. Second, compare asymptotics as $\lambda \to \infty$, where a unique slowest-decaying active band exists. Third, observe that this forces a band with the same centre but width $s/\sqrt{r}$ (cloud-free case) or $s/\sqrt{2}$ (cloudy case), which contradicts injectivity of the centres unless $r = 1$.

2. **Single-species density identifiability** (`DensityProfilesIdentifiableCts`, non-proportional cross-sections). This needs Abel inversion together with uniqueness for Laplace-type transforms, and probably a monotonicity hypothesis.

3. **Relaxing the structural assumptions** of Table 1: temperature-dependent cross-sections, an active background (which by Theorem 7 removes the profile degeneracy at the level of densities), refraction, and variable gravity.

Each target is stated as a Lean proposition, so a proof or a counterexample from a human or an automated prover is checked mechanically.

11 Conclusions

Formalising transmission-spectrum forward models makes exact degeneracies into theorems with explicit witnesses, and makes identifiability claims precise. In the exact model, the temperature profile is entirely unconstrained when the cross-sections are temperature independent and the background is invisible. That freedom vanishes once the composition is closed. In the analytic model, three of seven parameters are identifiable. The remaining four directions are an explicit gauge group, and its (R_0, P_0) part is broken, by an exponentially small amount, once the optically thick approximation is dropped. Between the two models, power rigidity of the opacity cone is the right notion for multi-absorber identifiability.

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Appendix: map from results to Lean

Result	Lean name	File
Lemma 1	`Atmosphere.speciesDensity_of_isothermal_isochemical`	`Assumptions`
Theorem 2	`Atmosphere.spectrum_eq_of_speciesDensity_eq`	`Atmosphere`
Theorem 3	`Atmosphere.temperature_profile_unidentifiable`	`ProfileDegeneracy`
Proposition 4	`Atmosphere.PhysicalVMR.retemper`	`ProfileDegeneracy`
Corollary 5, Theorem 6	`Atmosphere.isothermal_twin`, `Atmosphere.exists_inverted_twin`	`ProfileExamples`
Theorem 7	`Atmosphere.T_eq_of_speciesDensity_eq`	`ProfileClosure`
Proposition 9	`Iso.transitRadius_referenceGauge`, `Iso.transitRadius_normalization`, `Iso.transitRadius_T_g`, `Iso.transitRadius_T_m`	`Analytic`
Lemma 10, Theorem 11	`logAffine_rigidity`, `Iso.transitRadius_fibre`	`Identifiability`
Section 4.1	`Iso.grey_temperature_unidentifiable`, `Iso.lecavelier_slope`, `Iso.powerLaw_slope_degeneracy`	`Degeneracies`
Theorems 17 and 18	`Iso.scaleHeight_eq_of_powerRigid`, `Iso.exists_powerRelated_twin`	`Rigidity`
Theorem 19	`powerRigid_of_unique`	`GaussianRigidity`
Theorem 20	`Iso.transitRadiusE1_referenceGauge`, `expIntegralE1_strictAntiOn`	`E1Gauge`
Section 7	`not_densityProfilesIdentifiableCts_of_proportional`	`DensityTargets`

Table 2: Results and their Lean counterparts (all in namespace `ExoLean`).

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